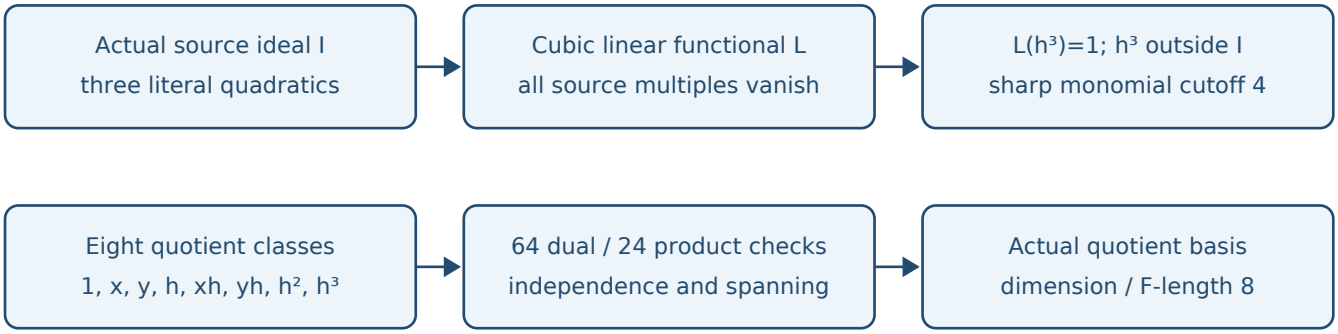
Exact algebra for the JV empty example

Item9 | 13 November 2026 | Whole gate 51/0 failures/0 trusted | [pcc:unverified]

The literal first Joux–Vitse toy system has an actual eight-dimensional homogeneous coordinate algebra over F101. Lean constructs its basis and proves that the previously established monomial cutoff of four is minimal. No Gröbner-basis oracle, generic regularity premise or native Hilbert output is assumed.



Actual objects. P=F101[x,y,h]; I is the ideal generated by the three printed coordinate quadratics after homogenization; A=P/I. The rows below are exact source coefficients.

	x^2	xy	y^2	xh	yh	h^2
f0	100	32	1	55	17	81
f1	29	34	0	71	83	16
f2	59	27	0	31	48	32

Lower cutoff. The genuine linear functional L uses weights (79,36,71,79,93,77,47,35,19,1) on $(x^3,x^2y,xy^2,y^3,x^2h,xyh,y^2h,xh^2,yh^2,h^3)$. For every polynomial g and each source generator f, L(gf)=0. Its full multiplication annihilator is an actual ideal containing I, while $L(h^3)=1$. Thus h^3 is outside I. Together with the earlier upper bound, every monomial of degree at least d belongs to I exactly when $d\geq 4$.

Exact dimension and support. Eight explicit dual-test polynomials give all 64 Kronecker-delta pairings with $(1,x,y,h,xh,yh,h^2,h^3)$, proving independence in the actual quotient. All 24 coordinate multiplication relations put their products back in the span. Polynomial induction and quotient surjectivity prove spanning. The actual basis gives finrank(F,A)=8 and Module.length(F,A)=8. Over every characteristic101 field, the source equations hold exactly at the origin: geometric support1, scalar-field length8.

Proof map. Namespace JVHilbertAudit: CubicDual.lean proves arbitrary-multiplier annihilation; SharpCutoff.lean proves source_hcube_not_mem and literal_jv_sharp_monomial_cutoff; DualBasis.lean proves basisVector_linearIndependent; QuotientBasis.lean constructs actualBasis and literal_jv_exact_algebra_certificate.

Assurance and limits. Lean4.35.0 / Mathlib56a005855ea17e0b0958f73ec801ae140b0a1679. Only propext, Classical.choice and Quot.sound. No holes, custom axioms, native_decide, raised elaboration limits or waived checks. Failed attempts are retained; local checks and independently signed standing are distinct. No new native experiment, paid compute or agent storage upload.

The length is explicitly as an F-module. This does not prove internal F4/solving degrees, generic Joux–Vitse Assumption2, a canonical graded Hilbert-function identification, source whole-curve order or scalar relations. Earlier issued ideal and source-field packets are unchanged. The source example is confirmed at the stated algebraic scope.

Sources. Joux–Vitse author manuscript p7, PDF SHA256 prefix6f3268a0ef8c; exact hash/read scopes in evidence/source-review.json. Prior ideal: research/jv_macaulay_certificate_20261113. Actual curve/descent: research/jv_source_field_audit_20261113. KB unavailable (MCP absent/wrapper127); bounded primary/repository/Mathlib fallback. No novelty claim. [pcc:unverified]